Tectonophysics, 98 (1983) 11–27
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

PALAEOMAGNETIC EVIDENCE FROM PELAGIC LIMESTONES FOR CLOCKWISE ROTATION OF THE IONIAN ZONE, WESTERN GREECE *

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(Received March 25, 1983)

ABSTRACT

Horner, F. and Freeman, R., 1983. Palaeomagnetic evidence from pelagic limestones for clockwise rotation of the Ionian zone, western Greece. In: E. McClelland Brown and J. VandenBerg (Editors), Palaeomagnetism of Orogenic Belts. *Tectonophysics*, 98: 11–27.

On the basis of a pilot study of all the major carbonate lithologies in the Ionian zone of Western Greece, the upper part of the Upper Senonian to Upper Eocene "Microbreccious Limestones" formation was selected as the most suitable facies for a detailed regional palaeomagnetic study. The 51 sites sampled are situated on flanks of anticlines or monoclines which post-date the acquisition of the characteristic remanent magnetization. The regional folding occurred during post-Late Eocene emplacement of nappe sheets which had undergone décollement on Triassic evaporitic horizons. The palaeomagnetic directions after correction for deviations of local fold axis orientation have a regional significance. After this correction the mean palaeodeclination is about 35°E relative to the geographic pole. With respect to the Early Tertiary African pole position, these palaeodeclinations indicate an apparent 38° clockwise rotation of the nappe sheets. The palaeoinclinations support a palaeogeographic model locating the carbonate sediments of the Ionian zone on the northeastern margin of the Adriatic Promontory of the African plate. The total rotation implied by the palaeodeclination data is possibly due to a combination of two different mechanisms:

(1) rotation of the nappe sheets during overthrust and emplacement during the main mid-Miocene compressional phase of deformation,

(2) rotation of the Ionian zone (basement and nappes) associated with extension in the Aegean basin during the last 13 m.y.

INTRODUCTION

Within the last few years many workers (Lowrie and Alvarez, 1975; VandenBerg et al., 1978; Channell et al., 1978) have used palaeomagnetic data from sedimentary rocks in Italy to provide important contributions to the growing body of geological

* Institut für Geophysik, ETH Zürich, Contribution Nr. 409.

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and geophysical data concerning the tectonic history of the peri-Adriatic region.

In contrast, Greece has until very recently been almost completely neglected from a palaeomagnetic point of view. Remanent magnetization of Pliocene Almopias volcanics from central Macedonia (Bobier, 1968) had negative polarity and expected directions, but the formation is too young for these results to be relevant for the earlier tectonic history. Directions in Upper Jurassic ophiolites indicated some 110° clockwise rotation of the Argolis peninsula relative to a sampling locality in the North Pindos area (Pucher et al., 1974). The reliability of the divergent Argolis and Pindos directions has been seriously questioned by Jacobshagen (1974) for geological and tectonic reasons. Awakened interest in this region is evidenced by the recent number of publications on the palaeomagnetic record from the Upper Miocene marine clay sequences from Western Crete (Langereis, 1979; Valet and Laj, 1981; Valente et al., 1982). During a study of the neotectonic features of the outer Aegean arc, Laj et al. (1982) collected Neogene samples from some of the Ionian and Aegean arc islands. They found evidence in sediments from Corfu of about 26° clockwise rotation with respect to the present geographic pole since the mid-Miocene. Data from Lower Oligocene to Upper Aquitanian Ionian flysch sections (Kissel et al., 1983) indicate a still earlier phase of some 30° additional clockwise rotation of Ionian zone sediments.

During the past three years, the entire Early Jurassic to Late Eocene carbonate sequence from the Ionian zone has been sampled at 130 sites providing some 2500 samples. The suitability of these different rock types for palaeomagnetic research has been evaluated as part of a comprehensive palaeomagnetic investigation. This report summarizes the data from white and pink cherty micrites of Early Tertiary age from 51 sites in the Ionian Zone of northwestern Greece. These rock types are equivalent to the Scaglia limestones from Italy which have proved to give reliable palaeomagnetic directions in several investigations (Lowrie and Alvarez, 1975; Channell et al., 1978; Lowrie et al., 1982).

TECTONIC SETTING OF THE IONIAN ZONE

The pronounced similarities between the Jurassic sequences of the Ionian zone and of the Marche–Umbrian region of central Italy were first noted by Renz (1923). The Mesozoic sedimentary rocks from the Ionian zone closely resemble those of the Central Apennines and the Southern Alps and are comparable to the sediments flanking the western margin of the Central Atlantic (Bernoulli, 1967, 1972; Bernoulli and Jenkyns, 1974).

The continuity of the South Tethyan carbonate facies from the Southern Apennines through the Dinarides and the Hellenides to the Taurides was used by Channell et al. (1979) as an additional argument for locating the palaeogeographic position of this region on the eastern margin of the Adriatic promontory of the African plate. Originally this margin had a quite different shape and position.

Intensive block-faulting during the Early Jurassic led to its differentiation into a number of elongate shallow marine platforms and deeper troughs which persisted until the Early Tertiary (Bernoulli and Renz, 1970). During the Late Eocene a marked change in depositional conditions occurred and extensive flysch deposits indicate the initiation of major tectonic activity to the east. As a consequence of the northward movement of Africa, the Adriatic plate collided with the Rhodope fragment. The timing of this collision is based on different interpretations of the geological and geophysical evidence and could have taken place over an extensive interval. Based on the best available geological data Burchfiel (1980) considers a Palaeogene age most likely. The Mesozoic–Tertiary sedimentary cover was detached from the basement, thrust westward along easily deformable Triassic evaporitic horizons and emplaced as a series of imbricated nappe sheets onto the stable Apulian platform. The major period of compression in the Ionian zone took place in the mid-Miocene (IGRS-IFP, 1966). The units comprising the Ionian zone are part of a plastically deforming continental margin affected by the tectonic development of the Aegean region (Tapponnier, 1977; Le Pichon and Angelier, 1979; Burchfiel, 1980).

GEOLOGY OF THE IONIAN ZONE

From central Thessaly and Attika westward to the Ionian Islands off the west coast of the Greek mainland, there is a varied sedimentary sequence of Mesozoic–Tertiary age, for which Aubouin (1959) derived his classical scheme of a geosyncline. He divided the geosynclinal belt into different "isopic zones" on the basis of their Mesozoic facies development. These zones provide a particularly well documented case of an orogen in which compressional features developed progressively in successive tectonic zones across the geosyncline through Cretaceous and Tertiary time, and in which major displacement of thrust sheets across the belts can be quantified.

In western Greece, the most external zones of the Hellenides are represented by a broad folded belt, running NNW–SSE along the coast of continental Greece and the Peloponnesus. From west to east the following zones have been distinguished (IGRS-IFP, 1966; see Fig. 1):

(a) the Apulian or Zante zone: neritic sedimentation from Triassic to Lower Miocene, represented in Apulia, Italy; found in a borehole on the southwestern Peloponnesus and on Zante island;

(b) the Preapulian or Paxos zone: sedimentation similar to (a), reaching into the Aquitanian, overlain by Miocene marl sequence, represented on Paxos island, western Levkas, and parts of the islands of Cephalonia and Zante;

(c) the Ionian zone: Triassic evaporites followed by an Upper Triassic to Upper Eocene limestone sequence, Upper Eocene to Aquitanian flysch, and transgressive Middle Miocene to Lower Pliocene marine sediments;

(d) the Gavrovo zone: reef and neritic, Upper Jurassic to Eocene limestones, covered by Upper Eocene flysch.

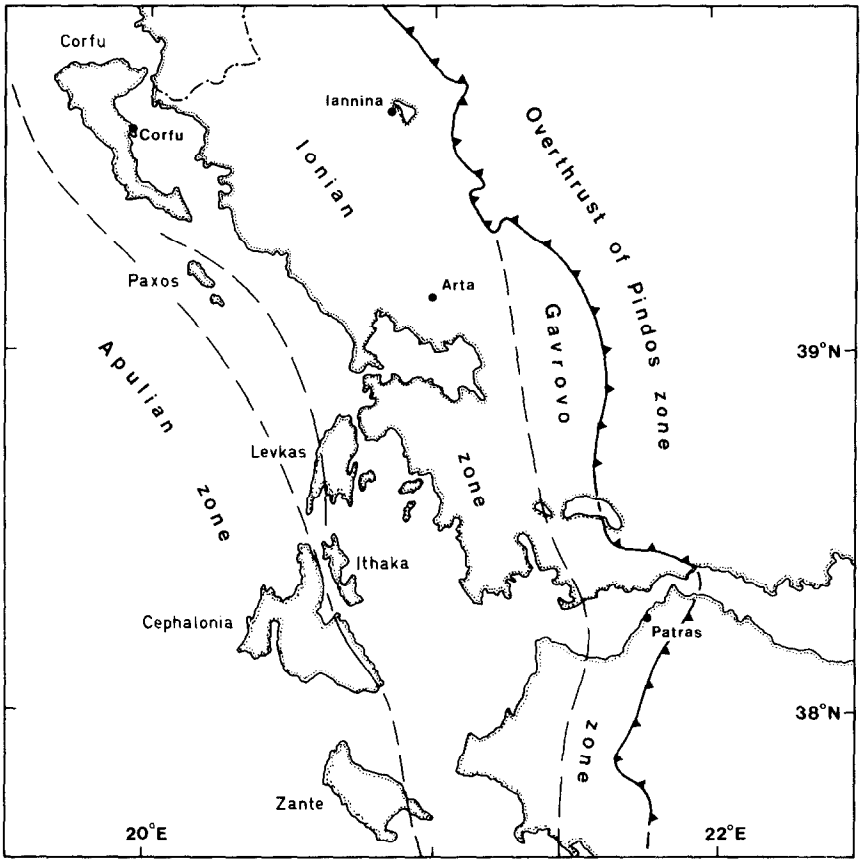


Fig. 1. External zones of the Hellenides.

LITHOLOGIES INVESTIGATED

From the Upper Tithonian onward, the abrupt lateral changes of facies and formation thicknesses characteristic for the Jurassic tended to be eliminated by overall higher sedimentation rates. The entire realm of the Ionian zone was covered by the monotonous facies of the Vigla limestone (Bernoulli and Renz, 1970). This formation is composed of well-bedded, white or light grey to yellowish fine-grained limestones, rich in planktonic organisms. It is characterized by bands and nodules of replacement chert and is comparable to the coeval pelagic sediments widespread over the whole of the Mediterranean Tethys. The Vigla limestone is partly equivalent to the Maiolica or Biancone formations of the Southern Alps and Central Apennines (Bernoulli, 1972).

In the central parts of the Ionian zone, a distinct phosphate horizon marks the upper limit of the Vigla and the transition to white and pink Upper Senonian to

Upper Eocene "Microbreccious Limestones" (Bernoulli and Renz, 1970). Lithologically, three limestone facies can be distinguished:

(1) massive calcarenites with molluscan debris and intraclasts of older limestones derived from the margins of adjacent carbonate platforms as lime turbidites;

(2) intercalated pelagic micritic limestones; and

(3) micritic limestones with a planktonic fauna.

The lower part of this series is characterized by facies (1) and (2) and is commonly called "Senonian Limestones" by Greek geologists. The upper part (3) is dominated by thin-bedded, very fine-grained ('porcellanous'), white and pink pelagic limestones, which often contain wavy and exotically shaped bands of chert. It is occasionally labeled "Sublithographic Limestones" (IGRS-IFP, 1966). This latter facies type is most suitable for palaeomagnetic purposes and supplied all of the cores for this paper. Micropalaeontological dating revealed Early Tertiary ages for all the samples of this facies.

It is occasionally difficult to make an unambiguous distinction between the similar facies of the upper part of the Vigla formation and the "Sublithographic Limestones". For the initial identification of lithofacies and age the 1 : 50,000 Geological Maps of Greece were consulted. In the field where recent road cuts exposed fresh outcrops, distinctive lithological characteristics used by Greek geologists served to differentiate between these two formations. In contrast to the uppermost part of the Vigla formation the upper part of the "Microbreccious Limestones" are characterized by chert lenses instead of chert bands, slightly thicker beds, more marly interbeds, and pronounced stylolites and laminations. To remove persistent uncertainty, all samples were routinely dated by micropalaeontological analysis of thin sections and washed samples.

CHOICE OF SAMPLING SITES

Faulted folds, diapiric structures, and low-angle thrusts, incurred during the Miocene–Pliocene tectonics, result in complex structures which had to be taken into account in the choice of suitable sites. The extensive Quaternary system of normal faults attests to the fact that Greece has long been one of the major active earthquake zones of the world. In addition, the Greek climate is conducive to the weathering of rocks. These factors combine to present the palaeomagnetist with a difficult task when searching for accessible outcrops fulfilling the above requirements in the Ionian zone. The fold axes should be horizontal or subhorizontal in order to avoid introducing artificial rotations when applying bedding correction. Even though corrections for plunging fold axes are available, these give a unique value only when the detailed folding history is known. Of particular importance for palaeomagnetic studies in orogenic belts are accurate determinations of the relationship between the strike of the local fold axes and the structural trend of the outcrop. Sampling was carried out only in those sites where the structural effects incurred

during emplacement and Quaternary block faulting were minimal. Regions where deformations are known to be due to evaporitic intrusions were avoided from the outset. In the Ionian zone, the lithology best satisfying both the structural and rock magnetic criteria is the upper part of the "Microbreccious Limestones" formation.

The initial aim was to sample extensive stratigraphic sequences in the different tectonic subunits of the Ionian zone. However, preliminary results of three stratigraphic sections covering the whole Ionian carbonate sequence raised serious doubts about the palaeomagnetic suitability of the Upper Tithonian to Lower Senonian Vigla formation and the lower part of the Upper Senonian to Upper Eocene "Microbreccious Limestones" formation. This was unfortunate since they represent a record of the tectonic history of the Adriatic region during the crucial Upper Cretaceous period. Although comparison of apparent polar wander paths from individual thrust sheets would have been more appropriate, the rock magnetic characteristics constrained the detailed sampling to lithologies which are palaeomagnetically more favorable.

REMANENT MAGNETIZATION

The remanent magnetization of the specimens was measured with a three-axis ScT cryogenic magnetometer interfaced with a HP-1000 minicomputer. The measuring configuration integrates a chosen number of readings (in general 20) per measurement and achieves a lower sensitivity limit of $6.0 \cdot 10^{-6}$ A/m. Replicate measurements (three or six) of each specimen overdefine the remanence vector, so that the accuracy of repeatability can be used as a rejection criterion at each stage of demagnetization (Lowrie et al., 1980).

During a detailed pilot study a large number of specimens from each lithology were demagnetized progressively in order to separate individual magnetization components of the natural remanent magnetization (N.R.M.) and to determine the suitability for palaeomagnetic purposes. Wherever possible, two specimens from the same sample were subjected to alternating field (A.F.) or thermal demagnetization. Progressive A.F. cleaning with peak alternating fields up to 80 mT was carried out in small (5–10 mT) steps using a two-axis tumbler; the earth's magnetic field was compensated with a Helmholtz coil system. Progressive thermal demagnetization involving 12–20 steps from room temperature to 700°C was carried out in air within a non-magnetic furnace (rest field less than 5 nT). Changes in magnetic susceptibility during progressive thermal demagnetization were monitored with a model KLY-1 bulk susceptibility bridge.

Acquisition of isothermal remanent magnetization (I.R.M.) in fields up to 4 T was used to identify the carriers of the remanence. Magnetic extracts were analyzed by electron microprobe and scanning electron microscope (S.E.M.). These observations were used to correlate specific magnetic mineral phases with components identified on demagnetization diagrams and I.R.M. acquisition curves.

The Scaglia-type fine-grained pelagic micrites of the "Sublithographic Limestones" appeared in cherty and non-cherty, white and pink varieties. In general the intensities varied inversely to the amount of chert present and proportionally to the degree of reddening. These two types had different rock magnetic characteristics and will therefore be considered separately.

Although weak, several non-cherty, white limestone varieties proved to be excellent recorders of the ancient magnetic field. Their natural remanences were characterized by two components: (1) a low unblocking temperature, very low coercivity phase which generally carries a recent field direction, and (2) a low coercivity, higher unblocking temperature phase which carries a stable endpoint characteristic remanent magnetization (Ch.R.M.). These characteristics are illustrated in the alternating field and thermal demagnetization curves of Fig. 2. The samples illustrated carry a viscous component with a recent field direction and a reversed Ch.R.M. component which can be clearly discerned and isolated (Fig. 2a, d). Unblocking temperatures between 500°C and 600°C (Fig. 2e), and moderate coercivities (Fig. 2b), indicate the presence of a relatively titanium-free magnetite component. A pronounced knee on the I.R.M. acquisition curves in fields of about 200 mT (Fig. 2c) results from the

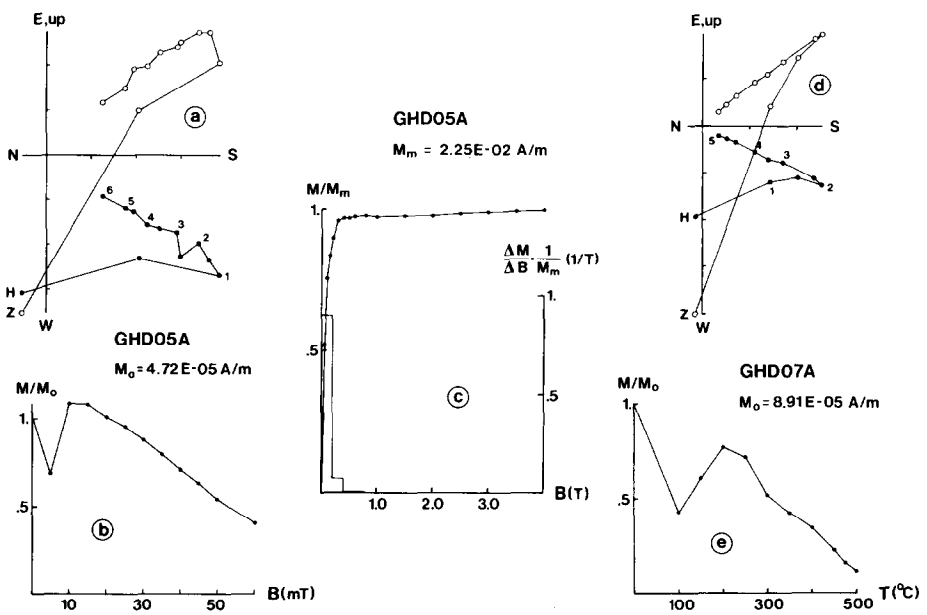


Fig. 2. Demagnetization diagrams in orthogonal projection (a, d), normalized intensity curves (b, e), and I.R.M. acquisition curve with coercivity spectra (c) of two characteristic specimens from white-colored "Sublithographic Limestones". Open and solid dots indicate components in the vertical and horizontal planes, respectively. Numbers on each vector diagram refer to peak A.F. demagnetizing field in tens of mT, or temperature in hundreds of °C. Unit on vector plots: (a) $1.0 \cdot 10^{-5}$ A/m; (d) $2.0 \cdot 10^{-5}$ A/m. Bedding corrected.

saturation of a magnetite phase. The two N.R.M. components may reflect different grain-size fractions of magnetite, the viscous component due to large multi-domain grains and the Ch.R.M. component due to smaller pseudo-single or single domain grains.

The pink varieties exhibit in general stronger initial N.R.M. and saturation I.R.M. intensities, higher susceptibilities and a more complex magnetic mineralogy than the white varieties. The characteristic features of reddish tint, high coercivities and unblocking temperatures over 600°C, indicate that pigment and specular haematite are present in significant amounts. Overgrowths on iron-titanium oxides and authigenic microcrystalline haematite are common features in magnetic extracts examined with the S.E.M. Since iron-oxide grains in the magnetic concentrate can oxidize during the laboratory extraction process, care must be taken to differentiate recent from ancient haematite. Freshly crystallized platelets and rosettes of haematite "rust" are distinguished by their perfect idiomorphic form, characteristic habit, lack of deformation, and bright red color under optical illumination. Ancient specular haematite is black under reflected light and can be observed before coating for S.E.M. work. In addition, compaction-deformed blades of microcrystalline haematite in the magnetic extract attest to their early diagenetic growth. These observations support a model of early diagenetic haematite growth and the acquisition of a

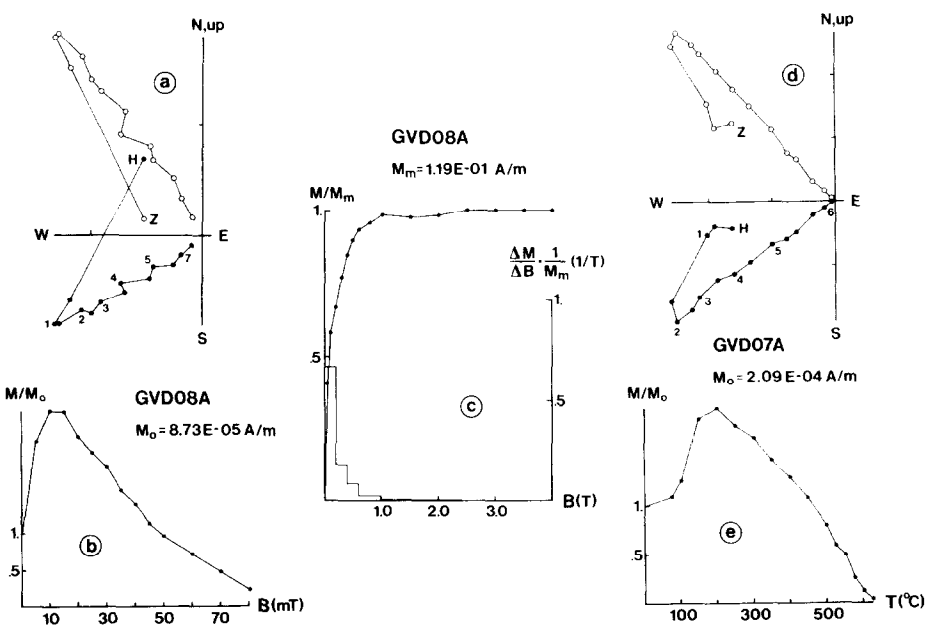


Fig. 3. Demagnetization diagrams in orthogonal projection (a, d), normalized intensity curves (b, e), and I.R.M. acquisition curve with coercivity spectra (c) of two representative specimens from pink-colored "Sublithographic Limestones". For symbols refer to Fig. 2. Unit on vector plots: $1.0 \cdot 10^{-4}$ A/m. Bedding corrected.

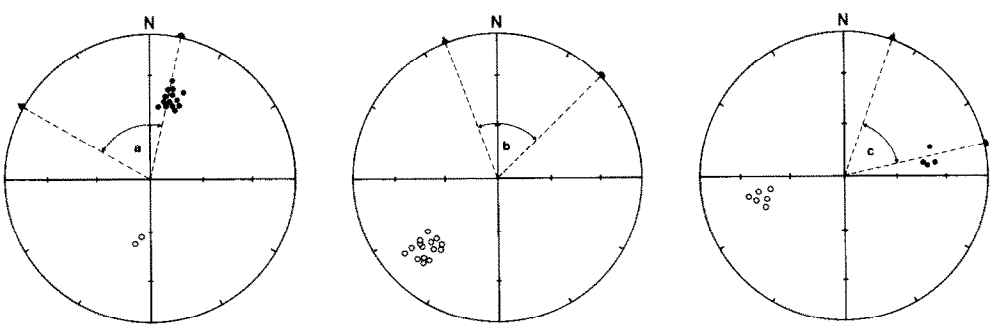
C.R.M. by haematite shortly after deposition as proposed by Channell et al. (1982) for the similar Scaglia Rossa lithology of the Umbrian Apennines.

As in the white varieties, however, a part of the remanent magnetization in the pink limestones is carried by a low coercivity and low unblocking temperature mineral (Fig. 3b, c), probably magnetite. Again a normal present field component overprinting a reversed Ch.R.M. is observable in the A.F. and thermal demagnetization curves of Fig. 3. A comparison of the I.R.M. curve in Fig. 2 with that of Fig. 3 shows that the knee in the curve, indicative of the approach to saturation, occurs at higher fields, ranging up to 1.0 T. This feature is due to the haematite component. As the spontaneous magnetization of haematite is roughly 0.5% that of magnetite, and the saturation isothermal remanence of the pink "Sublithographic Limestones" can be five times greater than that of the white variety, it is clear that the haematite component dominates the magnetic mineralogy.

PALAEOMAGNETIC RESULTS

Since thermal demagnetization consistently isolated better defined directions than A.F. demagnetization, thermal treatment was preferred for the routine measurement of the bulk of the samples. These were subjected to a partial progressive demagnetization treatment with at least four temperature steps based upon the behavior of the pilot samples of the particular lithology. From these demagnetization paths the direction of the characteristic remanent magnetization component was derived by three-dimensional least squares fitting, using a principal component analysis program modified after Kirschvink (1980).

The palaeomagnetic directions after correction for simple tectonic (bedding) tilt



Site KPA (L.Eocene) a=72° Site GVD (L.Paleocene) b=67° Site AC (M.Eocene) c=58°

Fig. 4. Equal-area projections of sample Ch.R.M. directions at three representative sites showing the relationship between site mean declination (solid half circle) and the strike of the respective local fold axis (solid triangle). Open and solid dots indicate average sample directions on upper and lower hemisphere respectively. Bedding corrected.

TABLE I

Summary of palaeomagnetic results *

Site	Age	Location		N/n	CDEC	CINC	BDEC	BINC	k_F	α_{95}
		(°N)	(°E)							
GHO	E. Tertiary	39.79	19.91	8/16	230.7	−48.3	228.6	−34.5	273.2	3.4
GHP	E. Tertiary	39.78	19.90	7/16	15.9	51.5	21.9	26.4	21.9	13.2
GHQ	E. Tertiary	39.78	19.90	5/5	48.9	65.0	40.0	37.8	26.7	14.2
GHR	E. Tertiary	39.78	19.90	9/9	25.7	67.4	29.4	39.5	143.3	4.3
LBA	E. Tertiary	38.67	20.69	8/8	36.1	−59.0	288.5	−61.0	14.7	14.9
LBB	E. Tertiary	38.67	20.69	6/9	221.5	72.2	86.1	54.6	38.2	11.0
ED	E. Palaeocene	39.79	19.91	10/20	34.1	46.1	19.1	41.7	80.3	5.4
GHM	L. Palaeocene	39.79	19.91	12/19	224.9	−51.3	224.4	−40.3	876.6	1.5
GHN	L. Palaeocene	39.79	19.91	8/9	232.0	−50.9	228.8	−43.4	373.3	2.9
GVD	L. Palaeocene	39.34	20.55	14/32	215.5	−72.8	233.2	−38.3	69.0	4.8
GVE	L. Palaeocene	39.34	20.55	16/34	204.4	−60.9	226.1	−31.1	108.1	3.6
AA	Eocene	38.89	21.14	11/11	259.9	−45.5	241.3	−45.7	67.4	5.3
GEM	E. Eocene	39.66	20.71	12/12	220.7	−41.2	217.2	−41.4	104.1	4.3
ASC	E. Eocene	39.82	19.98	17/17	186.0	−57.0	222.2	−29.5	63.9	4.5
EC	E. Eocene	39.58	20.98	13/20	207.8	−40.8	193.9	−52.8	45.9	6.2
GEA	E. Eocene	39.14	21.01	13/13	248.5	−79.2	232.7	−43.7	95.6	4.3
GEB	E. Eocene	39.12	21.04	17/14	18.1	65.3	42.7	38.5	54.0	4.9
GEC	E. Eocene	39.10	21.05	26/14	10.4	74.3	49.3	42.9	51.1	4.0
GEUB	E. Eocene	39.59	20.50	8/8	63.7	38.0	72.9	48.0	141.2	4.5
GEUA	E. Eocene	39.59	20.50	9/9	76.3	35.1	87.2	41.1	153.4	4.3
GHD	E. Eocene	39.79	19.93	13/32	212.3	−50.1	212.0	−35.1	402.9	2.1
GHI	E. Eocene	39.53	20.34	12/42	213.7	−29.2	224.5	−29.9	554.4	1.8
GHK	E. Eocene	39.53	20.34	12/36	225.5	−29.8	236.1	−27.6	110.8	4.1
GHT	E. Eocene	39.95	20.53	10/10	194.2	−54.9	207.9	−39.2	109.3	4.6
GHU	E. Eocene	39.95	20.53	10/10	185.9	−67.6	212.2	−53.5	124.8	4.3
GHV	E. Eocene	39.94	20.53	12/12	157.0	−61.7	185.4	−52.6	29.7	8.1

GSA	E. Eocene	39.23	20.64	20/30	236.2	−14.2	225.4	−34.8	42.5	5.1
GSB	E. Eocene	39.53	20.34	10/14	215.2	−28.2	225.2	−29.4	57.7	6.4
KPB	E. Eocene	39.68	20.33	10/24	36.4	−48.1	201.6	−29.4	123.4	2.7
AC	M. Eocene	38.53	21.09	10/10	103.5	9.3	78.3	40.3	215.7	3.3
AD	M. Eocene	38.53	21.09	12/12	205.5	−13.9	262.6	−45.9	145.3	4.3
EB	M. Eocene	39.58	20.97	12/13	234.2	−27.1	231.8	−40.0	41.4	6.8
GEG	M. Eocene	39.26	20.98	13/15	225.4	−58.4	246.3	−38.9	146.1	3.2
GEK	M. Eocene	39.56	20.79	12/15	249.1	−47.4	245.8	−44.9	68.7	4.6
GSC	M. Eocene	39.53	20.35	10/34	198.2	−42.1	223.5	−37.1	64.1	5.0
EA	L. Eocene	39.58	20.97	12/14	64.4	39.9	61.4	49.9	92.4	4.5
EGA	L. Eocene	39.23	20.98	9/16	208.6	−57.1	239.2	−43.3	730.6	1.9
ETA	L. Eocene	39.37	20.70	8/8	2.6	57.2	47.7	43.7	745.3	2.0
ETB	L. Eocene	39.37	20.70	9/11	358.3	59.7	51.7	47.5	953.0	1.7
ETC	L. Eocene	39.37	20.70	8/9	7.1	51.0	41.4	38.5	658.8	2.2
GEH	L. Eocene	37.30	20.77	14/16	62.1	70.2	66.8	44.4	110.3	3.5
GEI	L. Eocene	39.29	20.77	13/15	70.8	78.4	71.7	45.6	58.5	5.0
GEL	L. Eocene	39.58	20.77	12/14	248.1	−39.6	249.5	−46.7	94.0	4.1
GEN	L. Eocene	39.43	20.91	13/14	223.2	−43.5	225.8	−40.7	197.4	2.8
GEO	L. Eocene	39.45	20.86	13/16	25.0	52.7	40.3	40.2	63.2	4.7
GEP	L. Eocene	39.47	20.88	12/12	1.5	43.6	26.1	47.1	192.1	3.1
GEQ	L. Eocene	39.46	20.89	13/14	24.7	45.0	30.3	47.7	187.8	2.9
GER	L. Eocene	39.48	20.89	9/9	214.8	−52.3	223.3	−50.6	614.2	2.1
GET	L. Eocene	39.67	20.95	12/12	205.8	−50.8	199.3	−55.4	28.9	8.2
GSD	L. Eocene	39.53	20.35	15/39	36.6	43.7	46.6	39.3	133.9	3.3
KPA	L. Eocene	39.68	20.33	16/16	210.5	39.4	12.7	44.9	120.2	3.4
Mean direction of all "Sublithographic Limestones" site means										
(a)	after bedding correction, before correction for strike of local fold axes, relative to North (see Fig. 5a)						47.1	43.5	25.0	4.1
(b)	after bedding correction, after correction for strike of local fold axes, relative to local fold axes (see Fig. 5b)						65.1	42.5	50.3	2.8
(c)	after bedding correction, after correction for strike of local fold axes, relative to regional trend (see Fig. 5c)						35.1	42.5	50.3	2.8

* N/n—number of samples/specimens measured; CDEC/CINC—declination/inclination before bedding correction; BDEC/BINC—declination/inclination after bedding correction; k_F—Fisher's (1953) precision parameter; α_{95}—Fisher's (1953) confidence angle.

at the 51 sites investigated are clearly related to local structural trends. Figure 4 shows the characteristic remanent directions at three representative sites plotted on equal area projections. Each data point represents a sample average value of up to three specimens. Mean directions for these three sites are in general well-defined with radii of the circle of 95% confidence (α_{95}) around 4° (cf. Table I). However, the mean directions vary considerably from site to site. Clearly, thrust related local rotation will distort the palaeomagnetic directions if they accurately record pre-folding directions. The strike of the local fold axis (triangles in Fig. 4) is indicated for each site. The angle between the site means and the strike of local folding is similar for these three sites and ranges from 58° to 72°.

Figure 5a shows all the site mean directions after tectonic correction. To examine the relationship between the site mean directions and the local fold axes, each site mean has been replotted relative to the respective local fold axis (*FA* in Fig. 5b). The predominant tectonic style is imbrication of overthrusted nappe sheets and therefore most of the sites are located on monoclines or anticlines where the fold axes dip subhorizontally. Since this local structural correction amounts to a rotation about a vertical pole, only the declinations are affected. After applying this correction, there is a significant improvement in the grouping of the site means which had a precision factor $k_F = 25.0$ before correction and $k_F = 50.3$ after correction (see Figs. 5a and 5b; and also Table I). This result supports the interpretation of the palaeomagnetic directions as predating the tectonic emplacement of the thrust sheets.

The accuracy with which the strike can be measured depends upon the dip, and is most exact for steeply dipping beds. An accuracy of 5° for the strike of steep beds translated into a larger uncertainty for shallow beds (e.g., it is 15° for a dip of 20°). If the local fold axis lay within the confidence limits of the strike of bedding, the site was accepted. For the 26 sites which passed this criterion the angle between site-mean palaeodeclination and local fold axis was again averaged. The mean

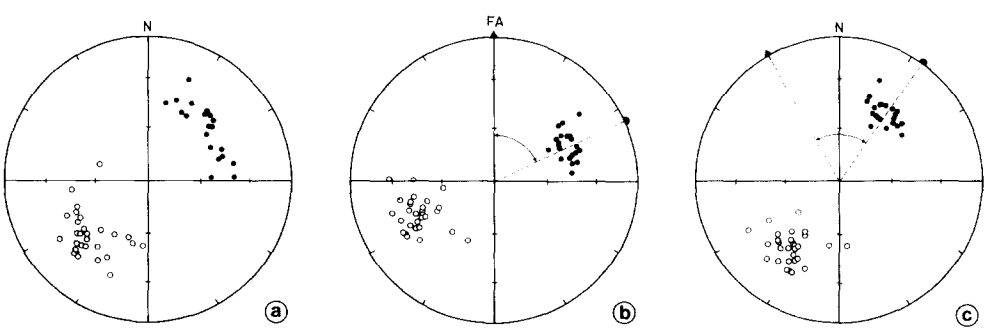


Fig. 5. All site means plotted on equal area projection (a) against geographic north; (b) relative to their respective local fold axes (*FA*); and (c) corrected for the regional strike (NNW) of the fold axes (solid triangle). Mean palaeodeclination is indicated by solid half circle. Open and solid dots represent site mean directions on upper and lower hemisphere respectively.

deviation of 67° was not significantly different from that of 65° for the complete set of 51 sites.

The regional strike of the folds is consistently NNW throughout the Dinarides and Hellenides and trends about 330°E in the Ionian zone of western Greece. To place the palaeomagnetic directions in a regional frame, the set must be rotated until the local fold axes strike along the regional trend (Fig. 5c). The resulting mean palaeomagnetic direction for the Early Tertiary "Sublithographic Limestones" from the Ionian Zone has then a declination of 35°E and an inclination of 43° (see Table I).

DISCUSSION

The palaeodeclinations depend very much on the tectonic history of the fold belt. These palaeomagnetic directions reflect the style of thrust folding for which there is extensive evidence in the Ionian zone. The dominant feature of the structural framework of the Ionian zone is low-angle thrust faulting. Although gravity sliding has been proposed as a possible structural driving force, suitable gradients do not seem to have been available and therefore large-scale westward gravity sliding in the Ionian zone is not usually considered as a likely thrust mechanism (Jenkins, 1972). Foreshortening of the basement may have taken place, and this could conceivably have affected the palaeomagnetic directions. However, there is at present a lack of knowledge concerning possible foreshortening of the basement but there is extensive evidence for large-scale overthrusting of the nappe sheets. In view of this situation, any serious models of basement movement based on palaeomagnetic results from the overlying sediments would be highly speculative.

The Ionian zone is part of a region where the major collisional effects between the African and Eurasian plates have been greatly modified by the formation of smaller continental fragments such as the Adriatic and the Rhodope plates and were subjected to complex regimes of extension and compression. The nappe sheets have been disrupted during emplacement and subsequently tectonized into a zone of continuous deformation characteristic of most of the Eastern Mediterranean (Tapponnier, 1977; Dewey and Sengör, 1979; Burchfiel, 1980). The interpretation of the clockwise rotation of the nappes of the Ionian zone is therefore beset with uncertainties.

Clockwise rotation of about 30° of the northwestern part of the outer Aegean arc around an Eulerian pole located in the southern Adriatic Sea has been proposed by Le Pichon and Angelier (1979). They associated the rotational movement with extension of the Aegean basin during the last 13 m.y. and challenged palaeomagnetists to test their idea. The part of the Ionian zone investigated in this study lies north of the assumed triple point marking the junction of the northwestern end of the Hellenic trench with the presumed western extension of the E–W trending Anatolian fault. As explained below, the palaeomagnetic results imply that the

Ionian zone should also be included in the area affected by the extension of the Aegean basin. Until now, the northern margin of Aegea has been identified with the "Karpenission–Lamia couloir". Le Pichon and Angelier (1979) have associated this series of E–W trending grabens with active normal faulting in the Gulf of Corinth and Peloponnesus region throughout the Plio-Quaternary. Significantly, they note that no major horizontal motion has occurred along this zone during the last 20 m.y. They reason that the westward movement of the Turkish plate along the Anatolian fault as well as the extension of Aegea has been taken up by a broad zone of deformation and conclude that Aegea therefore cannot be considered as a rigid plate.

Recent palaeomagnetic data from Corfu indicate a 26°-clockwise rotation with respect to the present geographic pole since the mid-Miocene (Laj et al., 1982) and therefore lend support to the model proposed by Le Pichon and Angelier (1979). Since there are no major geological or tectonic discontinuities between the mainland of northwestern Greece (Epirus) and the island of Corfu, results from both may be considered as representing the Ionian zone in general. Rotation of the Ionian zone sediments therefore implies that the region north of Peloponnesus could be also affected by the extensional tectonics associated with the evolution of the Hellenic arc and trench system since the Pliocene.

As part of the eastern margin of the Adriatic promontory of the African plate the Palaeocene–Eocene sediments of the Ionian zone should be expected to have a palaeodirection of 357° declination and 43° inclination. This calculation, based on the Early Tertiary African pole position of Van der Voo and French (1974), assumes a representative site location of 39.5°N, 20.5°E. The observed inclinations agree well with the predicted value (cf. Table I). After correcting for the effect of thrust related local fold axes rotations, the observed declinations imply a 38° rotation since the Early Tertiary. This interpretation is relevant only to the extent that the sediments are considered to be attached to the basement.

These Early Tertiary directions differ significantly from the 26° clockwise rotation established by the Neogene Corfuan data (Laj et al., 1982). The latter were calculated relative to the geographic pole which is virtually identical to the African pole for the Pliocene. This difference implies an additional phase of clockwise rotation of some 10° between the Early Tertiary and the Pliocene.

A recent palaeomagnetic study of 16 flysch sites of Lower Oligocene to Upper Aquitanian age in Epirus and Akarnania (Kissel et al., 1983) shows a total clockwise rotation of some 60° with respect to the geographic pole since the Oligo-Miocene. This is considerably larger than the 38° we propose. Direct comparison with our Early Tertiary data from carbonate beds is at the moment uncertain because Kissel et al. (1983) considered that corrections to their flysch data to allow for local folding structures were unnecessary. However, these two studies bring palaeomagnetic evidence from different lithologies which agree on an additional resultant clockwise rotation between the Early Tertiary and the mid-Miocene.

This earlier phase of rotation could be due to back-arc spreading associated with volcanic activity in the Aegean Sea as proposed by Kissel et al. (1983). Another mechanism could be the compressive deformation of the fold belts as proposed by Tapponnier (1977). He considered that the curvature of the Alpine orogenic belt might be due to "rigid-plastic" deformation of the continental margins during the northeasterly impingement of the Adriatic promontory. Although Tapponnier's model applies primarily to basement deformation, it would have had the effect of rotating the allochthonous external basin sediments, originally located on opposite sides of the Adriatic promontory, in opposite directions. Sedimentary basins lying on the western margin (for example, the present Marche–Umbrian region of central Italy) were rotated counterclockwise. This rotation is evident in the Umbrian palaeomagnetic directions (Lowrie and Alvarez, 1975; VandenBerg et al., 1978; Channell et al., 1978). Similarly, the folded belts of the sedimentary basins lying originally on the eastern margin (the present Ionian zone) would be rotated clockwise during collision and emplacement as the data from this study suggest. If the Early Tertiary palaeomagnetic data from the Ionian zone are used to rotate the region back to its original attitude relative to the Early Tertiary African pole, the strike of the fold belts is then about 290°. This palaeogeographic trend aligns better with the Umbrian basins as could be expected from geological considerations. This model provides a possible explanation for the pre-Pliocene rotation of the Ionian zone that is not accounted for by the model of post-Miocene extension proposed by Le Pichon and Angelier (1979).

ACKNOWLEDGEMENTS

We thank W. Lowrie and F. Heller for critically reading the manuscript. C.T. Klootwijk and an anonymous reviewer supplied a number of much appreciated comments, some of which have been integrated into the final version. M. Baltuck and J.G. Ogg helped initially with sample collection. We thank the Institute of Geological and Mining research (IGME), Athens, for providing the necessary permissions, S. Papastavrou for geological advice, J.P. Beckmann and M. Toumarkine for palaeontological dating, and H.U. Nissen and P. Waegli for SEM assistance. Financial support was provided by the Swiss National Science Grant 2.415-0.79.

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